9. Compare Linear and Polynomial Regression using Python

Aim

To compare the performance of **Linear Regression and Polynomial Regression** using Python.

Software Required

- Python 3.x
- NumPy
- Scikit-learn
- Matplotlib

Procedure

1. Create a nonlinear dataset.
2. Split the data into training and testing sets.
3. Train a Linear Regression model.
4. Transform the input using Polynomial Features.
5. Train the Polynomial Regression model.
6. Calculate the Mean Squared Error for both models.
7. Compare their performance.

Code

```
import numpy as np

from sklearn.linear_model import LinearRegression

from sklearn.preprocessing import PolynomialFeatures

from sklearn.metrics import mean_squared_error

X = np.array([1, 2, 3, 4, 5, 6]).reshape(-1, 1)

y = np.array([1, 4, 9, 16, 25, 36])

# Linear Regression

linear = LinearRegression()

linear.fit(X, y)

pred_linear = linear.predict(X)

# Polynomial Regression
```

```
poly = PolynomialFeatures(degree=2)

X_poly = poly.fit_transform(X)

polynomial = LinearRegression()

polynomial.fit(X_poly, y)

pred_poly = polynomial.predict(X_poly)

print("Linear Regression MSE:",

    mean_squared_error(y, pred_linear))

print("Polynomial Regression MSE:",

    mean_squared_error(y, pred_poly))
```

Output:

```
print("Polynomial Regression MSE:",
      mean_squared_error(y, pred_poly))
```

```
•••  Linear Regression MSE: 6.222222222222218
     Polynomial Regression MSE: 4.3126450477355994e-29
```

Result

Thus, **Polynomial Regression** performed better than Linear Regression for the given nonlinear dataset.